

EXAMEN CALCUL DIFERENTIAL SI INTEGRAL SERIA 13

OFICIU: **1 punct**

SUBIECTUL 1. (2 puncte)

Sa se studieze natura seriei $\sum_{n=1}^{+\infty} \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} \cdot \frac{1}{n^2}$.

SUBIECTUL 2. (2 puncte)

Sa se determine punctele de extrem local ale functiei $f : \mathbb{R}^2 \rightarrow \mathbb{R}, f(x, y) = x^4 - 16xy + y^4 \forall (x, y) \in \mathbb{R}^2$.

SUBIECTUL 3. (2 puncte)

Sa se studieze convergenta simpla si uniforma a sirului de functii $f_n : [5, +\infty) \rightarrow \mathbb{R}, f_n(x) = \frac{nx^2}{n+x} \forall x \in [5, +\infty), \forall n \in \mathbb{N}$.

SUBIECTUL 4. (3 puncte)

a) Sa se calculeze $\iint_D x^2 y dx dy$, unde $D = \{(x, y) \in \mathbb{R}^2 \mid y \leq -2x + 2, x \geq 0, y \geq 0\}$.

b) Fie $f : [a, b] \rightarrow \mathbb{R}$ o functie de clasa C^1 pe $[a, b]$. Sa se demonstreze ca $\lim_{n \rightarrow \infty} \int_a^b f(x) \sin(nx) dx = 0$.

Examen calcul diferențial și integral

Carpințanu
Alexandru
Grupa 13.1

Exercițiul 1

$$\sum_{n=1}^{+\infty} \frac{1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1)}{2 \cdot 4 \cdot 6 \cdot \dots \cdot (2n)} \cdot \frac{1}{n^2}$$

$$x_n = \frac{1 \cdot 3 \cdot \dots \cdot (2n-1)}{2 \cdot 4 \cdot 6 \cdot \dots \cdot 2n} \cdot \frac{1}{n^2}, \quad \forall n \in \mathbb{N}^+$$

$$\lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n} = \lim_{n \rightarrow \infty} \frac{1 \cdot 3 \cdot \dots \cdot (2n+1)}{2 \cdot 4 \cdot \dots \cdot 2(n+1)} \cdot \frac{2 \cdot 4 \cdot \dots \cdot 2n}{1 \cdot 3 \cdot \dots \cdot (2n-1)} \cdot \frac{n^2}{(n+1)^2}$$

$$= \lim_{n \rightarrow \infty} \frac{n^2 (2n+1)}{(n+1)^2 (2n+2)} = \lim_{n \rightarrow \infty} \frac{2n^3 + n^2}{(n^2 + 2n + 1)(2n + 2)}$$

$$= \lim_{n \rightarrow \infty} \frac{2n^3 + n^2}{2n^3 + 2n^2 + 4n^2 + 4n + 2n + 2} = 1$$

$$\lim_{n \rightarrow \infty} n \left(\frac{x_n}{x_{n+1}} - 1 \right) = \lim_{n \rightarrow \infty} n \left(\frac{1 \cdot 3 \cdot \dots \cdot (2n-1)}{2 \cdot 4 \cdot \dots \cdot 2n} \cdot \frac{2 \cdot 4 \cdot \dots \cdot 2(n+1)}{1 \cdot 3 \cdot \dots \cdot (2n+1)} \cdot \frac{(n+1)^2}{n^2} - 1 \right)$$

$$= \lim_{n \rightarrow \infty} n \left(\frac{(2n+2)(n+1)^2}{(2n+1)n^2} - 1 \right)$$

$$= \lim_{n \rightarrow \infty} n \left(\frac{(2n+2)(n+1)^2 - (2n+1)n^2}{(2n+1)n^2} \right)$$

$$\lim_{n \rightarrow \infty} n \cdot \frac{(2n+2)(n^2+2n+1) - 2n^3 - n^2}{2n^3 + n^2}$$

$$= \lim_{n \rightarrow \infty} n \cdot \frac{2n^3 + 4n^2 + 2n + 2n^2 + 4n + 2 - 2n^3 - n^2}{2n^3 + n^2}$$

$$= \lim_{n \rightarrow \infty} n \cdot \frac{5n^2 + 6n + 2}{2n^3 + n^2} = \lim_{n \rightarrow \infty} \frac{5n^2 + 6n + 2}{2n^2 + n} = \frac{5}{2} > 1$$

$\Rightarrow \sum_{n=1}^{+\infty} \frac{1 \cdot 3 \cdot \dots \cdot (2n-1)}{2 \cdot 4 \cdot 6 \cdot \dots \cdot 2n} \cdot \frac{1}{n^2}$ este convergentă
 Cał. Riesz-Duhamel

2/ $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x, y) = x^4 - 16xy + y^4$ ($\forall$) $x, y \in \mathbb{R}^2$
 f funcție continuă pe $\mathbb{R}^2 \Rightarrow Df = \emptyset$

~~... (scrierea răsturnată a funcției) ...~~

$\frac{\partial f}{\partial x}(x, y) = 4x^3 - 16y$ ($\forall$) $x, y \in \mathbb{R}^2$
 $\frac{\partial f}{\partial y}(x, y) = -16x + 4y^3$ ($\forall$) $x, y \in \mathbb{R}^2$

$\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ funcții cont. pe $\mathbb{R}^2$
 $\mathbb{R}^2$ multime deschisă

$\Rightarrow f$ funcție diferentiaabilă pe $\mathbb{R}^2$
 $Df = \emptyset$

$$\begin{cases} \frac{\partial f}{\partial x}(x, y) = 0 \\ \frac{\partial f}{\partial y}(x, y) = 0 \end{cases} \quad \text{po } \mathbb{R}^2$$

$$\begin{cases} 4x^3 - 16y = 0 \quad | :x \\ 4x^3 - 16x = 0 \quad | :x \end{cases} = \begin{cases} 4x^2 - 16y = 0 \\ 4x^2 - 16x = 0 \end{cases}$$

(-)

$$4x^2 - 4x^2 - 16x + 16x = 0$$

$$\Rightarrow 4(x^2 - y^2) = 0$$

$$\Rightarrow (x^2 - y^2) = 0 \Rightarrow (x^2 - y^2)(x^2 + y^2) = 0$$

$$\Rightarrow (x - y)(x + y)(x^2 + y^2) = 0$$

Case 1: $(x - y) = 0 \Rightarrow x = y$

Case 2: $x + y = 0 \Rightarrow x = -y$

Case 3: $x^2 + y^2 = 0 \Rightarrow x = y = 0 \Rightarrow x_1 = 0$
 $y_1 = 0$

Case 1: $x = y$

$$\Rightarrow 4x^3 - 16x = 0$$

$$x(4x^2 - 16) = 0$$

$$x(x^2 - 4) = 0$$

$$x_1 = 0$$

$$(x - 2)(x + 2) = 0$$

$$x_2 = 2 \Rightarrow -y_2 = 2$$

$$x_3 = -2 \Rightarrow -y_3 = -2$$

Case 2: $x = -y$

$$4y^3 - (16 \cdot (-y)) = 0$$

$$4y^3 + 16y = 0$$

$$y(y^2 + 4) = 0$$

$$y_1 = 0$$

$$y^2 + 4 = 0$$

$$y^2 = -4 \quad \text{no real roots.}$$

$$(0,0), (0,2), (0,-2), (2,0), (2,2), (2,-2), (-2,0), (-2,2), (-2,-2) \in \mathbb{R}^2$$

$$\Rightarrow C = \{(0,0), (0,2), (0,-2), (2,0), (2,2), (2,-2), (-2,0), (-2,2), (-2,-2)\}$$

$$\frac{\partial^2 f}{\partial x^2}(x,y) = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right)(x,y) = (4x^3 - 16y)'_x = 12x^2 \quad (\forall (x,y) \in \mathbb{R}^2)$$

$$\frac{\partial^2 f}{\partial x \partial y}(x,y) = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right)(x,y) = (4y^3 - 16x)'_x = -16 \quad (\forall (x,y) \in \mathbb{R}^2)$$

$$\frac{\partial^2 f}{\partial y \partial x}(x,y) = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right)(x,y) = (4x^3 - 16y)'_y = -16 \quad (\forall (x,y) \in \mathbb{R}^2)$$

$$\frac{\partial^2 f}{\partial y^2}(x,y) = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right)(x,y) = (4y^3 - 16x)'_y = 12y^2$$

$$\frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial x \partial y}, \frac{\partial^2 f}{\partial y \partial x}, \frac{\partial^2 f}{\partial y^2} \text{ fct. cont pe } \mathbb{R}^2$$

$\mathbb{R}^2$ multime deschisă

$$\Rightarrow f \text{ fct. diferențiabilă de ordin } 2 \text{ pe } \mathbb{R}^2 \Rightarrow D_2 = \emptyset$$

$$H_f(0,0) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2}(0,0) & \frac{\partial^2 f}{\partial x \partial y}(0,0) \\ \frac{\partial^2 f}{\partial y \partial x}(0,0) & \frac{\partial^2 f}{\partial y^2}(0,0) \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -16 \\ -16 & 0 \end{pmatrix}$$

~~Qu~~ $\Delta_1 = 0$

$\Delta_2 = -256 \Rightarrow (0,0)$ nu este pct de extrem local

$$H_f(0,2) = \begin{pmatrix} 0 & -16 \\ -16 & 12 \cdot 2^2 \end{pmatrix}$$

$\Delta_1 = 0$

$\Delta_2 = -256 \Rightarrow (0,2)$ nu este pct de extrem local

$$H_f(0,-2) = \begin{pmatrix} 0 & -16 \\ -16 & 12 \cdot (-2)^2 \end{pmatrix}$$

$\Delta_1 = 0$

$\Delta_2 = -256 \Rightarrow (0,-2)$ nu e pct de extrem local

$$H_f(2,0) = \begin{pmatrix} 12 \cdot 2^2 & -16 \\ -16 & 0 \end{pmatrix}$$

$\Delta_1 = 12 \cdot 4 = 48 \geq 0$

$\Delta_2 = -256 \leq 0$

$\Rightarrow (2,0)$ nu e pct de extrem local

$$H_f(2,2) = \begin{pmatrix} 12 \cdot 2^2 & -16 \\ -16 & 12 \cdot 2^2 \end{pmatrix}$$

$$\Delta_1 = 48 > 0$$

$$\Delta_2 = 48^2 - (-16)^2 = 48^2 - 16^2 > 0$$

Cr. Sylvester

$\Rightarrow (2,2)$ este pct de minim local al fct. f

$$H_f(2,-2) = \begin{pmatrix} 12 \cdot 2^2 & -16 \\ -16 & 12 \cdot (-2)^2 \end{pmatrix}$$

$$\Delta_1 = 48 > 0$$

$$\Delta_2 = 48^2 - 16^2 > 0$$

Cr. Sylvester

$\Rightarrow (2,-2)$ este pct de minim local al fct. f

$$H_f(-2,0)$$

$$= \begin{pmatrix} 12 \cdot (-2)^2 & -16 \\ -16 & 0 \end{pmatrix}$$

$$\Delta_1 = 48 > 0$$

$$\Delta_2 = -256 < 0$$

$\Rightarrow (-2,0)$ nu este pct de extrem local

$$H_f(-2,2) = \begin{pmatrix} 12 \cdot (-2)^2 & -16 \\ -16 & 12 \cdot 2^2 \end{pmatrix}$$

$$\Delta_1 = 48 > 0$$

$$\Delta_2 = 48^2 - 16^2 > 0$$

Cr. Sylvester

$\Rightarrow (-2,2)$ este pct de min. local al fct. f

$$H_f(-2, -2) = \begin{pmatrix} 12 - (-2)^2 & -16 \\ -16 & 12 - (-2)^2 \end{pmatrix}$$

$$\Delta_1 = 48 > 0$$

$$\Delta_2 = 48^2 - 16^2 > 0$$

Cr. Sylvester

$\Rightarrow (-2, -2)$ este pt de min local al fct. f

$$\Delta_3 = \{(2, 2), (2, -2), (-2, 2), (-2, -2)\}$$

$$\Delta_4 = \emptyset$$

$\Rightarrow$ punctele de extrem local ale lui f sunt $(2, 2), (2, -2), (-2, 2), (-2, -2)$

$$3/ f_n: [\bar{s}, +\infty) \rightarrow \mathbb{R}$$

$$f_n(x) = \frac{nx^2}{n+x} \quad (\forall) x \in [\bar{s}, +\infty), (\forall) n \in \mathbb{N}$$

$$\text{Fie } x \in [\bar{s}, +\infty)$$

$$\lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \frac{nx^2}{n+x} = x^2 \quad (\forall) x \in [\bar{s}, +\infty)$$

$$A = [\bar{s}, +\infty)$$

$$f_n: A \rightarrow \mathbb{R}$$

$$f: A \rightarrow \mathbb{R}, f(x) = x^2 \quad (\forall) x \in A$$

$$f_n \xrightarrow{[\bar{s}, +\infty)} f$$

Fix $n \in \mathbb{N}^+$

$$\sup_{x \in [\delta, +\infty)} |f_n(x) - f(x)| = \sup_{x \in [\delta, +\infty)} \left| \frac{n+x^2}{n+x} - \frac{n+x}{x^2} \right|$$

$$= \sup_{x \in [\delta, +\infty)} \left| \frac{n x^2 - n x^2 - x^3}{n+x} \right| = \sup_{x \in [\delta, +\infty)} \frac{x^3}{n+x}$$

$$x = n \in [\delta, +\infty)$$

$$\Rightarrow \sup_{x \in [\delta, +\infty)} \frac{x^3}{n+x} \geq \frac{n^3}{n+n} = \frac{n^3}{2n} = \frac{n^2}{2}$$

$$\Rightarrow \sup_{x \in [\delta, +\infty)} |f_n(x) - f(x)| \geq \frac{n^2}{2} \quad (\forall) n \in \mathbb{N}^+$$

$$\Rightarrow \lim_{n \rightarrow \infty} \sup_{x \in [\delta, +\infty)} |f_n(x) - f(x)| = +\infty$$

$$\Rightarrow \lim_{n \rightarrow \infty} \sup_{x \in [\delta, +\infty)} |f_n(x) - f(x)| \neq 0$$

$$\Rightarrow f_n \not\rightarrow f \text{ on } [\delta, +\infty)$$

$$4/a \iint_D x^2 y \, dx \, dy$$

$$D = \{ (x, y) \in \mathbb{R}^2 \mid y \leq -2x + 2, x \geq 0, y \geq 0 \}$$

$$0 \leq y \leq -2x + 2$$

$$\Rightarrow -2x + 2 \geq 0$$

$$\Rightarrow 2x \leq 2 \Rightarrow x \leq 1 \quad \begin{matrix} x \geq 0 \\ \hline \Rightarrow x \in [0, 1] \end{matrix}$$

$$\Rightarrow D = \{ (x, y) \in \mathbb{R}^2 \mid x \in [0, 1], y \leq -2x + 2 \}$$

$[0, 1] \subseteq \mathbb{R}$ interval compact in $\mathbb{R} \Rightarrow [0, 1] \in \mathcal{J}(\mathbb{R})$

Fie $\alpha, \beta : [0, 1] \rightarrow \mathbb{R}$

$$\alpha(x) = 0$$

$$\beta(x) = -2x + 2$$

$$(\forall) x \in [0, 1]$$

α, β fct. cont pe $[0, 1]$

D multime simplă în raport cu axa $Ox \Rightarrow D \in \mathcal{J}(\mathbb{R}^2)$

Fie $f : D \rightarrow \mathbb{R}$

$$f(x, y) = x^2 y$$

f fct. cont pe D

D multime simplă în raport cu axa Ox

$\Rightarrow f$ fct. integrabilă Riemann pe D

$$\begin{aligned}
\iint_D f(x, y) dx dy &= \int_0^1 \left(\int_{\alpha(x)}^{\beta(x)} f(x, y) dy \right) dx \\
&= \int_0^1 \left(\int_0^{-2x+2} x^2 - y dy \right) dx = \int_0^1 \left(\frac{y^2 x^2}{2} \Big|_0^{-2x+2} \right) dx \\
&= \int_0^1 \frac{x^2}{2} ((-2x+2)^2 - 0) dx \\
&= \int_0^1 \frac{x^2}{2} \cdot (4x^2 - 4x + 4) dx \\
&= \frac{1}{2} \int_0^1 4x^4 - 4x^3 + 4x^2 dx \\
&= 2 \int_0^1 x^4 - x^3 + x^2 dx \\
&= 2 \left(\frac{x^5}{5} \Big|_0^1 - \frac{x^4}{4} \Big|_0^1 + \frac{x^3}{3} \Big|_0^1 \right) \\
&= 2 \cdot \left(\frac{1}{5} - \frac{1}{4} + \frac{1}{3} \right) \\
&= 2 \cdot \left(\frac{12 - 5 + 20}{60} \right) \\
&= 2 \cdot \frac{-3 + 20}{30} = \frac{17}{30}
\end{aligned}$$